

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

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REG NO : 192524317

COURSE CODE: CSA0708

COURSE OUTCOME COVERED

CO5:

Measure network performance using key metrics with simulation tools (BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 Total Marks:30 Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)			) Proficient g (2)	
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well structured sub problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

ANSWERS

1. Internet Connectivity for a Remote Branch Office

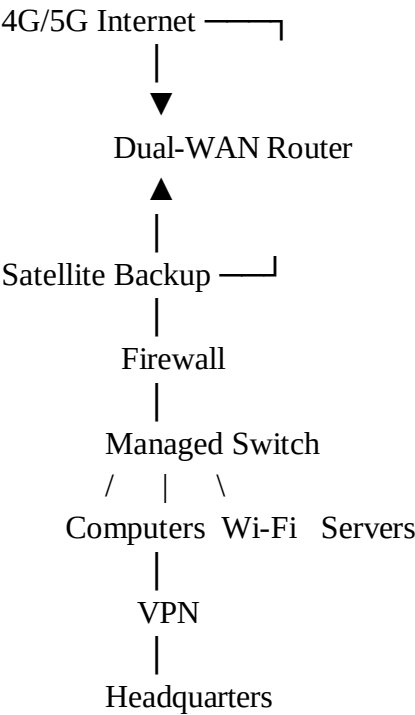
Recommended Internet Connection

Since high-speed wired internet is unavailable, the best solution is to use 4G/5G Fixed Wireless Internet as the primary connection and Satellite Internet as a backup connection. 4G/5G provides good speed for cloud applications, email, and video conferencing. Satellite internet can maintain connectivity when the mobile network fails. A Dual-WAN router should automatically switch to the backup connection during failure.

Required Networking Devices

- 4G/5G Router with external antenna
- Satellite modem for backup
- Dual-WAN router
- Managed switch
- Wireless Access Points
- Next-Generation Firewall
- UPS for power backup

Network Architecture



Security Mechanisms

A Site-to-Site VPN should be used for secure communication between the branch office and headquarters. A Next-Generation Firewall protects the network from unauthorized access and cyberattacks.

VLANs can separate:

- Employee network
- Server network
- Guest Wi-Fi

QoS should prioritize important traffic such as video conferencing, VPN, and cloud **applications**.

Justification

4G/5G is cheaper and easier to deploy than installing fiber in a rural location. The satellite

connection improves reliability by acting as a backup. The Dual-WAN router ensures continuous operation through automatic failover.

Final Recommendation:

4G/5G Primary + Satellite Backup + Dual-WAN Router + Firewall + VPN + VLAN + QoS provides a good balance of bandwidth, reliability, security, cost, and future scalability.

2. Wireless Network Design for a University

Requirement

The university needs to provide Wi-Fi access for more than **3,000 students** across classrooms, laboratories, libraries, and hostels. The network should support many simultaneous users while reducing interference and keeping costs low.

Access Point Placement

Since only a limited number of Access Points are available, they should be installed based on **user density**.

High-Density Areas

More Access Points should be placed in:

- Libraries
- Large classrooms
- Laboratories
- Lecture halls
- Hostels

Fewer APs can be placed in corridors and low-traffic areas. A **wireless site survey** should be conducted before installation to identify dead zones and sources of interference.

Frequency Band Selection

Frequency Band	Advantages	Recommended Use
2.4 GHz	Wider coverage and better wall penetration	Older devices and large coverage areas
5 GHz	Higher speed and less interference	Students, classrooms, labs, and high-density areas

The 5 GHz band should be preferred because it provides better speed and more channels. The 2.4 GHz band can be used where wider coverage is required.

Channel Planning

Access Points should use different non-overlapping channels to reduce interference.

For example:

AP 1 → Channel 1

AP 2 → Channel 6

AP 3 → Channel 11

This helps reduce signal interference between nearby APs.

Security and Authentication

The recommended authentication method is:

- **WPA3-Enterprise**
- **802.1X Authentication**
- **RADIUS Server**

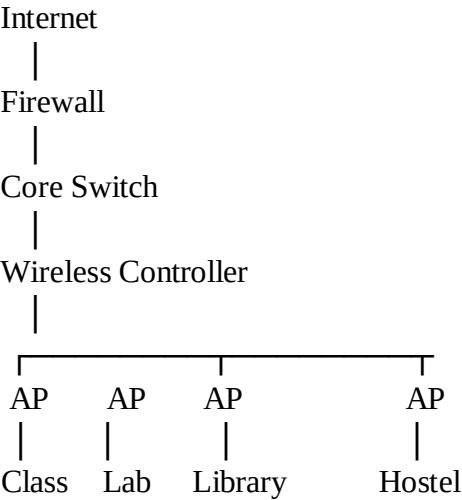
Different users should be separated using VLANs:

- Student VLAN
- Faculty VLAN

- Administration VLAN
- Guest VLAN

Guest users should only receive internet access and should not access university systems.

Simple Architecture



Justification

Placing Access Points according to user density reduces the total number of APs and lowers cost. Using 5 GHz improves performance and reduces interference. WPA3-Enterprise and 802.1X provide secure authentication.

Final Recommendation:

Use centrally managed Wi-Fi 6 Access Points, density-based placement, 5 GHz priority, proper channel planning, VLANs, and WPA3-Enterprise authentication. This balances coverage, performance, security, and cost.

3. Secure Network Architecture for a Financial Institution

Requirement

A financial institution handles sensitive customer information, banking transactions, and employee data. Therefore, the network must provide strong cybersecurity while maintaining good performance and controlling operational costs. The recommended approach is a layered security architecture.

Network Segmentation

The network should be divided into different security zones using VLANs and firewalls.

Recommended Segments

1. **DMZ** – Public web servers and online banking services
2. **Employee VLAN** – Employee computers
3. **Application VLAN** – Banking applications
4. **Database VLAN** – Customer and transaction data
5. **Admin VLAN** – Network and system administrators
6. **Guest VLAN** – Visitors with internet-only access

This prevents an attacker or unauthorized user from easily moving between different parts of the network.

Firewall Deployment

Two levels of firewall protection should be used.

Perimeter Firewall

Placed between the internet and the organization's network.

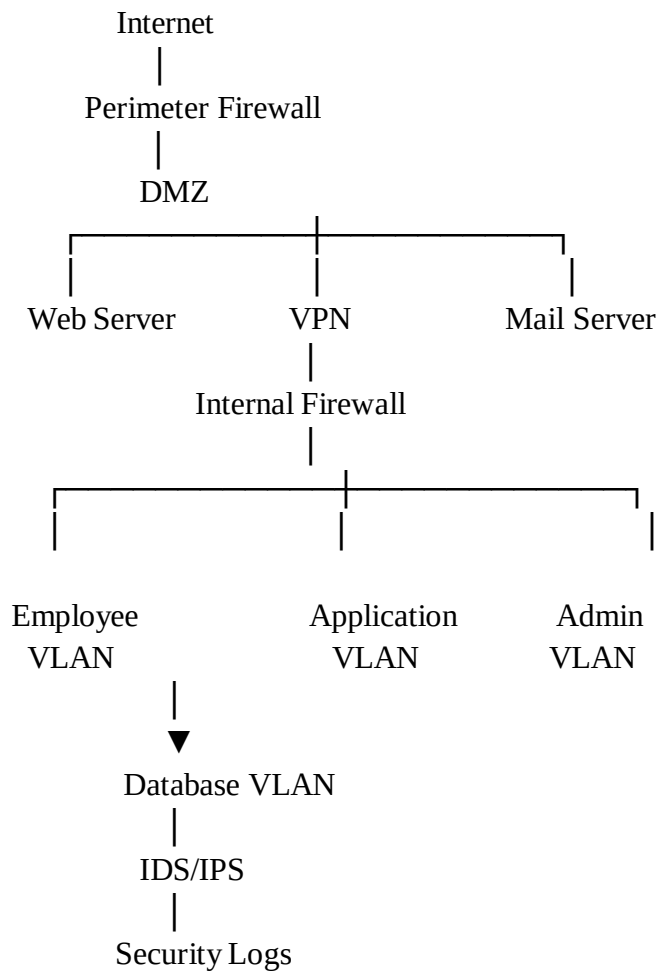
It protects against:

- Unauthorized access
- Malware
- External attacks
- Suspicious traffic
-

Internal Firewall

Placed between important internal networks such as the application and database networks. This provides additional protection for sensitive customer information.

Architecture



Authentication and Access Control

The institution should use **Multi-Factor Authentication (MFA)**.

For example:

Username + Password

+

OTP / Authenticator App

=

Secure Login

Role-Based Access Control (RBAC) should also be implemented. Employees should only access the systems and data required for their job.

For example:

- Teller → Transaction system only
- Manager → Reports and approvals
- Database Administrator → Database management
- IT Administrator → Network management

This follows the **Principle of Least Privilege**.

Intrusion Detection and Prevention

The organization should deploy:

- **IDS** – Detects suspicious network activity
- **IPS** – Detects and automatically blocks malicious traffic

A centralized **SIEM or log management system** can collect logs from:

- Firewalls
- Servers
- IDS/IPS
- Authentication systems

This helps security administrators detect attacks and investigate security incidents.

Secure Remote Access

Employees working remotely should access the internal network through:

VPN + Multi-Factor Authentication

This encrypts communication and prevents unauthorized access.

Cost and Performance Considerations

To control costs, the organization can use:

- VLANs instead of completely separate physical networks
- Centralized authentication
- Centralized security monitoring
- Strategically placed IDS/IPS
- QoS for important banking applications

This reduces unnecessary infrastructure costs while maintaining strong protection.

Final Recommendation

The financial institution should use:

Network Segmentation + VLANs + DMZ + Perimeter Firewall + Internal Firewall + MFA + RBAC + VPN + IDS/IPS + Centralized Logging

This architecture provides:

- **Strong security** for customer information
- **Good network performance**
- **Support for compliance requirements**
- **Controlled operational cost**
- **Future scalability**

Overall Conclusion

Scenario	Best Solution
Remote Branch Office	4G/5G + Satellite Backup + VPN + Firewall
University Wi-Fi	Wi-Fi 6 + Proper AP Placement + 5 GHz + WPA3
Financial Institution	VLANs + DMZ + Firewalls + MFA + IDS/IPS